

Matthew Tm Week 4



Q1. Claim: if $\vdash_{SS} \phi$, then $\models_{SS} \phi$

We use 2 lemmas:

Lemma 1: SS -axioms are SS -valid i.e.

$$PL1 \models_{SS} \phi \rightarrow (\psi \rightarrow \phi)$$

$$PL2 \models_{SS} (\phi \rightarrow (\psi \rightarrow \chi)) \rightarrow ((\phi \rightarrow \psi) \rightarrow (\phi \rightarrow \chi))$$

$$PL3 \models_{SS} (\neg \psi \rightarrow \neg \phi) \rightarrow ((\neg \psi \rightarrow \phi) \rightarrow \psi)$$

$$K \models_{SS} \Box(\phi \rightarrow \psi) \rightarrow (\Box \phi \rightarrow \Box \psi)$$

$$T \models_{SS} \Box \phi \rightarrow \phi$$

$$SS \models_{SS} \Diamond \Box \phi \rightarrow \Box \phi$$

Proof:

$$\bullet PL1: \models_{SS} \phi \rightarrow (\psi \rightarrow \phi)$$

$$\text{AFC that } \exists I \text{ st } V_I(\phi \rightarrow (\psi \rightarrow \phi)) = 0$$

$$\therefore V_I(\phi) = 1 \text{ \& } V_I(\psi \rightarrow \phi) = 0$$

$$\text{but then } V_I(\psi) = 1 \text{ \& } V_I(\phi) = 0 \text{ \& } \therefore$$

You have to do this in a

Kripke frame, not a model
for propositional logic.

i.e. use $V_w(\phi \rightarrow (\psi \rightarrow \phi), w) =$

0, not $V_I(\phi \rightarrow (\psi \rightarrow \phi)) = 0$,

$$\bullet PL2: \models_{SS} (\phi \rightarrow (\psi \rightarrow \chi)) \rightarrow ((\phi \rightarrow \psi) \rightarrow (\phi \rightarrow \chi))$$

$$\text{AFC } \exists I \text{ st } V_I((\phi \rightarrow (\psi \rightarrow \chi)) \rightarrow ((\phi \rightarrow \psi) \rightarrow (\phi \rightarrow \chi))) = 0 \text{ and so on}$$

$$\therefore V_I(\phi \rightarrow (\psi \rightarrow \chi)) = 1, V_I((\phi \rightarrow \psi) \rightarrow (\phi \rightarrow \chi)) = 0$$

$$\therefore V_I(\phi \rightarrow \psi) = 1, V_I(\phi \rightarrow \chi) = 0$$

$$\therefore V_I(\phi) = 1, V_I(\chi) = 0$$

$$\therefore V_I(\psi) = 1$$

$$\therefore V_I(\psi \rightarrow \chi) = 0, \therefore V_I(\phi \rightarrow (\psi \rightarrow \chi)) = 0 \text{ \& } \therefore$$

$$\bullet PL3: \models_{SS} (\neg \psi \rightarrow \neg \phi) \rightarrow ((\neg \psi \rightarrow \phi) \rightarrow \psi)$$

$$\text{AFC that } V_I(\neg \psi \rightarrow \neg \phi) \rightarrow ((\neg \psi \rightarrow \phi) \rightarrow \psi) = 0$$

$$\therefore V_I(\neg \psi \rightarrow \neg \phi) = 1, V_I(\neg \psi \rightarrow \phi) \rightarrow \psi = 0$$

$$\therefore V_I(\neg \psi \rightarrow \phi) = 1, V_I(\psi) = 0$$

$$\therefore V_I(\neg \psi) = 1, \therefore V_I(\phi) = 1 \Rightarrow V_I(\neg \phi) = 0$$

$$\bullet SS: \models_{SS} \Diamond \Box \phi \rightarrow \Box \phi$$

AFC that there is an SS-model st $V_I(\Diamond \Box \phi \rightarrow \Box \phi, u) = 0$

$$\therefore V_I(\Diamond \Box \phi, u) = 1, V_I(\Box \phi, w) = 0$$

$$\therefore V_I(\Box \phi, v) = 1 \text{ for some } \therefore V_I(\phi, u) = 0 \text{ for some}$$

$$\Downarrow \quad v \in U \text{ st } Ruv$$

$$u \in U \text{ st } Rvu$$

$$\therefore V_I(\phi, v') = 1 \text{ for all}$$

$$v' \in U \text{ st } Rvv'$$

Due to symmetry, $Ruv \Rightarrow Rvu$

By transitivity, $Ruv \& Rvu \Rightarrow Rvu$

so the world u in which $V_I(\phi, u) = 0$ is accessible from world v i.e. Rvu

since $V_I(\phi, v') = 1$ for all $v' \in U$ st Rvv'

choose $v' = u$ st. $V_I(\phi, u) = 1$ $\times$

Lemma 1 proven

Lemma 2: SS-rules preserve SS validity i.e.

MP: if $\models_{SS} \phi$ & $\models_{SS} \phi \rightarrow \psi$, then $\models_{SS} \psi$

NEC if $\models_{SS} \phi$, then $\models_{SS} \Box \phi$

Proof. • MP: if $\models_{SS} \phi$ & $\models_{SS} \phi \rightarrow \psi$, then $\models_{SS} \psi$

$$\text{if } \models_{SS} \phi, \& \models_{SS} \phi \rightarrow \psi, \\ \text{then } \models_{SS} \psi$$

then $V_M(\phi, u) = 1, V_M(\phi \rightarrow \psi, u) = 1$ for every $u \in U$ & for every M You should be a list

Then, in any model M & world u , if $V_M(\phi, u) = 1$, more careful here. To prove

then $V_M(\psi, u) = 1$ too

$$\therefore \models_{SS} \psi$$

$\models_{SS} \psi$ start with, fix M and w , and so use (1) and (2) to get $V_M(\psi, w) = 1$

• NEC: if $\vdash_{SS} \phi$, then $\vdash_{SS} \Box \phi$

Assume $\vdash_{SS} \phi$. Then $\forall M, \forall u \in W$ s.t. that $M, V_M(\phi, u) = 1$

Then since for any M , $V_M(\Box \phi, u) = 1 \quad \forall u \in W$

$\therefore V_M(\Box \phi, u) = 1$. Since this fact holds for all M ,
 $\Box \phi$ is valid in SS

i.e. $\vdash_{SS} \Box \phi$ 

Lemma 2 proven

Now, we use induction to show that the final line of a SS -proof sequence with n rule applications is SS -valid

Claim. $\vdash_n \phi$ means that there is a proof of ϕ (in SS) in which there is n application of SS -rules, MP or NEC.
Thus, $\vdash_n \phi$ implies $\vdash_{SS} \phi$

Proof. We prove this by strong induction on n i.e. length of proof.

Base Case. $n=0$

Suppose $\vdash_0 \phi$. By def, ϕ is an SS -axiom. By lemma 1, $\vdash_{SS} \phi$

Let n be given

IH: Suppose for $m < n$: $\vdash_m \phi$ implies $\vdash_{SS} \phi$

We wts that if $\vdash_n \phi$, then $\vdash_{SS} \phi$ (given the IH)

Suppose $\vdash_n \phi$. Then, the last line ϕ occurs in three cases:

(1) ϕ is an SS -axiom

$\therefore$ By lemma 1, $\vdash_{SS} \phi$

(2) ϕ is obtained by NEC

Then $\phi \models \mathcal{D}\psi$, and $\vdash_m \psi$, with $m < n$

By IH, $\vdash_{ss} \psi$,

$\therefore \vdash_{ss} \mathcal{D}\psi$ by Lemma 2, and $\vdash_{ss} \phi$

ϕ is obtained by MP

Then $\vdash_{m_1} \psi$, $\vdash_{m_2} \psi \rightarrow \phi$ with $m_1, m_2 < n$

By IH, $\vdash_{ss} \psi$, & $\vdash_{ss} \psi \rightarrow \phi$

by Lemma 2, $\vdash_k \phi$

$\therefore$ for any proof of length n , if $\vdash_{ss} \phi$, then $\vdash_{ss} \phi$

Q2 (a) Claim: if $\Gamma \vdash_s \delta$ and $\Sigma, \delta \vdash \phi$, then $\Gamma, \Sigma \vdash_s \phi$

Proof. Assume $\Gamma \vdash_s \delta$ and $\Sigma, \delta \vdash \phi$. So, there is a proof A of δ from Γ and a proof B of ϕ from Σ and δ . Let C be the result of concatenating the proofs A & B in that order. So, C begins with formulas of A , then followed by B . Then, the last iff of C is the last iff of B , ϕ . We want to show that C counts as a proof i.e. that each of C is either an axiom, a member of Γ or of Σ or follows from an earlier line by an S-rule. Since A is, by assumption, a proof of δ from Γ , the formula on any line j for all lines from A is either an axiom, a member of Γ or follows from an earlier line $i < j$ by an S-rule. Since B is a proof from Σ, δ , the formula on any line k here is either (i) an axiom, (ii) a member of Σ , (iii) δ , or (iv) follows from an earlier line by an S-rule. (i), (ii), (iv) fulfill the criteria for C to count as a proof — (iii) does because δ occurred in the last line of the proof A . So δ is either an axiom, a member of Γ or followed from an earlier line by an S-rule. The last part is okay because any line earlier than the last line of A is also earlier than whatever line δ follows from by an S-rule. So, C is a valid proof as any iff on any line is either (i) an axiom, (ii) a member of Γ, Σ or (iii) follows from an earlier line by an S-rule.

We can deduce Cut by easily generalizing Cut 1

Claim (Cut): if $\Gamma, \delta_1, \dots, \Gamma_n \vdash_s \delta_n$ and $\Sigma, \delta_1, \dots, \delta_n \vdash \phi$, then $\Gamma, \Gamma_1, \dots, \Gamma_n, \Sigma \vdash_s \phi$

Proof: Notice that we can generalize the proof in the following way. We let A_i be a proof of S_i from T_i , $i=1, \dots, n$, and B be a proof of ϕ from $\Sigma, S_1, \dots, S_n$. As per the above, let C be the concatenation of $A_1, \dots, A_n \vee B$, in the following order. We call the part of C that is $A_1, \dots, A_n$ the first phase $\vee$ the second phase. We want to show that any line j in C is either: (i) an axiom, (ii) a member of $T_1, \dots, T_n, \Sigma$, (iii) follows from an earlier line $k < j$ by a Σ -rule. What we agreed about A is the same for $A_1, \dots, A_n$. So any line from the concatenated proof in the first phase satisfies (i)-(iii). In the second phase, like the proof for cut 1, we only have to show that any of the S_i 's satisfies (i)-(iii). They clearly do, since S_i is the last line of corresponding proof A_i which means S_i is either an axiom, a member of T_i or follows from an earlier line via Σ -rule. Again, any line earlier than the last line of A_i is also earlier than line S_i .
 $\therefore$ Any line in the proof of C is (i)-(iii). Claim established.

You have to be careful, here to attempted to prove Cut from (b). The Deduction Theorem is scratch, but the question asked you to prove it.

Claim: If $T, \phi \vdash_{PL} \psi$, then $T \vdash_{PL} \phi \rightarrow \psi$ using Cut 1. You only used your own proof of Cut 1. It's a small but subtle difference.

Proof. Suppose $T, \phi \vdash_{PL} \psi$. Thus, there exists a proof A from T, ϕ to ψ . Each line α_i is either a member of T or ϕ , an axiom, or follows from earlier lines of proof by MP. The last line of A is ψ . We will prove a stronger claim for each α_i in proof A , $T \vdash_{PL} \phi \rightarrow \alpha_i$.

and DT obviously follows since $\alpha_n = \psi$ for any proof of length n .

We prove this by induction on the length of ϕ 's proof.

BC: Consider the first line α_1 . Then we have 2 cases for α_1 . Either:

1. α_1 is an axiom. Then $T \vdash \phi \rightarrow \alpha_1$ since

1. α_1 Axiom
 2. $\phi \rightarrow \alpha_1$ PL1, I, MP

2. α_1 is a premise from the set $T \cup \{\phi\}$. Then, by dichotomy, either α_1 is ϕ or not ϕ .

(a) if α_1 is ϕ , $T \vdash \phi \rightarrow \alpha_1$ since $\vdash \phi \rightarrow \phi$ is a PL-theorem.

(b) if α_1 is not ϕ , $T \vdash \phi \rightarrow \alpha_1$, as well since

1. α_1 Premise

2. $\phi \rightarrow \alpha_1$, PL1, I, MP

IH: If $\forall n (n < m) \vdash \phi \rightarrow \alpha_n$, then $\vdash \phi \rightarrow \alpha_m$

So, we assume that all earlier lines have the property that $\vdash \phi \rightarrow \alpha_n$. Thus, suppose that an follows from earlier lines in the proof by MP. This means that earlier lines have the forms $\chi \rightarrow \alpha_m$ and χ . By the IH, $\vdash \phi \rightarrow (\chi \rightarrow \alpha_m)$ & $\vdash \phi \rightarrow \chi$. Then, $\phi \rightarrow \chi, \phi \rightarrow (\chi \rightarrow \alpha_m) \vdash \phi \rightarrow \alpha_m$ since

1. $\phi \rightarrow \chi$ Premise ✓
2. $\phi \rightarrow (\chi \rightarrow \alpha_m)$ Premise
3. $(\phi \rightarrow (\chi \rightarrow \alpha_m)) \rightarrow ((\phi \rightarrow \chi) \rightarrow (\phi \rightarrow \alpha_m))$ PL2
4. $(\phi \rightarrow \chi) \rightarrow (\phi \rightarrow \alpha_m)$ MP, 2, 3
5. $\phi \rightarrow \alpha_m$ MP1, 4

By Cut, $\vdash \phi \rightarrow \alpha_m$

Intermediate claim established

DT falls out as a corollary as established $\nparallel$

Q3 Def [K-admissibility]. A rule R is K-admissible iff $\vdash_K \psi$ holds whenever $\vdash_K \phi_1$ and ... and $\vdash_K \phi_n$ i.e. if $\vdash_K \phi_1$ and ... and $\vdash_K \phi_n$, $\vdash_K \psi$.

(a) Claim: $\frac{\phi \rightarrow \psi}{\Box \phi \rightarrow \Box \psi}$ Becker is K-admissible, where $\Box$ represents some finite string of $\Diamond$ s and $\Box$ s.

Proof. We assume that $\vdash_K \phi \rightarrow \psi$. Then, we want to show that $\vdash_K \Box \phi \rightarrow \Box \psi$ follows.

We prove this on induction on the length of $\Box$.

BC: We have two cases. Either $\Box \phi \rightarrow \Box \psi$ or $\Diamond \phi \rightarrow \Diamond \psi$

1) $\vdash_K \Box \phi \rightarrow \Box \psi$. Since we know that there is a proof of $\phi \rightarrow \psi$

1. $\phi \rightarrow \psi$ By hypothesis ✓

2. $\Box(\phi \rightarrow \psi)$ NEC

3. $\Box(\phi \rightarrow \psi) \rightarrow (\Box \phi \rightarrow \Box \psi)$ K

4. $\Box \phi \rightarrow \Box \psi$ MP, 2, 3

2. $\vdash_K \Diamond \phi \rightarrow \Diamond \psi$. Since we know that there is a proof of $\phi \rightarrow \psi$

1. $\phi \rightarrow \psi$ By hypothesis

2. $\psi \rightarrow \sim \phi$ PL (contradiction)

3. $\Box \psi \rightarrow \Box \sim \phi$ NEC, K, MP

4. $\sim \Box \psi \rightarrow \sim \Box \sim \psi$ PL contradiction

✓

IH: if $\vdash_k \Box_n \phi \rightarrow \Box_n \psi$, then $\vdash_k \Box_{n+1} \phi \rightarrow \Box_{n+1} \psi$

Assume $\vdash_k \Box_n \phi \rightarrow \Box_n \psi$. $\therefore$ There is a proof of $\Box_n \phi \rightarrow \Box_n \psi$ from $\phi \rightarrow \psi$

Now, $\Box_{n+1} = \Box \Box_n$ or $\Diamond \Box_n$. We have shown by the Base Case that $\vdash_k \Box \Box_n \phi \rightarrow \Box \Box_n \psi$ and $\vdash_k \Diamond \Box_n \phi \rightarrow \Diamond \Box_n \psi$ holds.

$\therefore \vdash_k \Box_{n+1} \phi \rightarrow \Box_{n+1} \psi$ holds

IH established, claim proven as length of $\Box$

✓

(b) Claim: $\frac{\chi(\alpha)}{\chi(\phi)}$ is k-admissible where $\chi(\phi)$ uniformly substitutes ϕ for α in $\chi(\alpha)$

Proof: Assume $\vdash_k \chi(\alpha)$. So, there exists a proof A of $\chi(\alpha)$. We want to show that $\vdash_k \chi(\phi)$

Then, any line in the proof of $\chi(\alpha)$ is either an axiom or follows from an earlier line by a S-rule (MP or NEC). Notice that we can replace α in all the lines j in A where α occurs with ϕ , since the lines in A that contain α are either an MPL instance of an axiom schema and thus any ϕ instead of α will still be an MPL-instance of that axiom, or follows from an MPL-instance of an earlier line by a S-rule. Since any line derived from an S-rule depends on another earlier line

that either is (i) an MPL-instance of an axiom, (ii) a line that followed from an MPL-instance of an axiom via an S-rule or (iii) a line that followed from an earlier line which followed from an S-rule, we can define any line that follows from some S-rule recursively

$\alpha_k = \psi(\vec{\alpha}_k)$, where $\vec{\alpha}_k$ is a vector of proof lines earlier than α_k

WLOG, we can define k as the distance of α_k from the earliest axiom line it depends on. So, any substitution of α for ϕ in the MPL-instances of all axioms will "transfer" via the S-rule. Hence, The last line $\chi(\alpha)$ will instead be $\chi(\phi)$ of the proof. $\therefore \vdash_k \chi(\phi)$ ✓

(c) Claim: if $\vdash_K \phi_1, \vdash_K \phi_2, \dots, \vdash_K \phi_n$ and $\vdash_{PL} (\phi_1 \rightarrow (\phi_2 \rightarrow \dots (\phi_n \rightarrow \psi)))$
 then $\vdash_K \psi$

Proof. Suppose that $\vdash_K \phi_1, \dots, \vdash_K \phi_n$. Then there are proofs $A_1, \dots, A_n$ of $\phi_1, \dots, \phi_n$ from $T_i = \emptyset, i=1, \dots, n$

Since $(\phi_1 \rightarrow \dots (\phi_n \rightarrow \psi))$ is a MPL tautology, then there is a proof B of $(\phi_1 \rightarrow \dots (\phi_n \rightarrow \psi))$. Then suppose we have a proof C of ψ that concatenates

$A_1, \dots, A_n, B$ in the first phase and a continuation 2nd phase with a proof D . Then, 2)
 for C to be a valid proof of ψ , each line must either be (i) axiom, (ii) follows from an earlier line via a S -rule (MP or NEC)

Since PL is complete

so $\vdash_{PL} (\phi_1 \rightarrow \dots (\phi_n \rightarrow \psi))$

then $\vdash_K (\phi_1 \rightarrow \dots (\phi_n \rightarrow \psi))$

Since any PL-theorem is

provable in K

$\vdash_K (\phi_1 \rightarrow \dots (\phi_n \rightarrow \psi))$

By Assumption, $A_1, \dots, A_n, B$ are all valid proofs since every line is either (i) or (ii).

Now, we must show that every line in D satisfies (i) or (ii).

This holds since ψ follows from a continuous application of MP on initially

the last line of B i.e. $(\phi_1 \rightarrow \dots (\phi_n \rightarrow \psi))$ and the last line of A_i

and then the line resulting from that use of MP and A_i , for i in $[2, n]$

$\therefore$ Every line in D is (ii) since the proof C is ordered by playing the D proof at the end. $\therefore \vdash_K \psi$

2) I am struggling to reach this line which crucially defines $\hookrightarrow$ and $\leftrightarrow$,
 (d) Claim: if $\vdash_K \phi_1 \leftrightarrow \phi_2$, then $\vdash_K \chi(\phi_1) \leftrightarrow \chi(\phi_2)$ but the rest looks right

Proof. Suppose $\vdash_K \phi_1 \leftrightarrow \phi_2$. Then, there is a proof A of $\phi_1 \leftrightarrow \phi_2$.

Furthermore, there is a proof A_1 of $\phi_1 \rightarrow \phi_2$ and a proof A_2 of $\phi_2 \rightarrow \phi_1$

where A is a concatenation of proofs A_1, A_2 , and an additional last line that yields $\phi_1 \leftrightarrow \phi_2$ by definition

Now notice that in each proof A_i , each line is either (i) an axiom or

(ii) follows from an earlier line via an S -rule. We want to show that our

new proof of $\chi(\phi_1) \leftrightarrow \chi(\phi_2)$, call this A' , and $\chi(\phi_2) \leftrightarrow \chi(\phi_1)$,

call this A'' , satisfies (i) and (ii) for all lines

We prove this by induction on the length of $\chi(\phi_i)$ reasoning within an already WLOG, consider any proof A_i existing proof, induction is a good

idea, try it in a different way

BC: Trivial since $\chi(\phi_1) = \phi_1, \chi(\phi_2) = \phi_2$

By assumption, $\vdash_K \chi(\phi_1) \leftrightarrow \chi(\phi_2)$ since $\vdash_K \phi_1 \leftrightarrow \phi_2$

I.H. Suppose for all $n \in \mathbb{N}$, $\vdash_k \chi_n(\phi_1) \leftrightarrow \chi_n(\phi_2)$

Then consider 3 cases:

$$1. \chi_n(\phi_1) = \sim \chi_{n-1}(\phi_1), \chi_n(\phi_2) = \sim \chi_{n-1}(\phi_2)$$

Then consider:

$\vdots$

$$k. \chi_{n-1}(\phi_1) \rightarrow \chi_{n-1}(\phi_2) \quad (\text{by IH})$$

$$k+1. \sim \chi_{n-1}(\phi_2) \rightarrow \sim \chi_{n-1}(\phi_1) \quad \text{PL (contradposition)}$$

$\vdots$

$$k+j. \chi_{n-1}(\phi_2) \rightarrow \chi_{n-1}(\phi_1) \quad (\text{by IH})$$

$$k+j+1. \sim \chi_{n-1}(\phi_1) \rightarrow \sim \chi_{n-1}(\phi_2) \quad \text{PL (contradposition)}$$

$$k+j+2. \sim \chi_{n-1}(\phi_1) \leftrightarrow \sim \chi_{n-1}(\phi_2) \quad \text{by def}$$

$$\text{So, } \vdash_k \sim \chi_{n-1}(\phi_1) \leftrightarrow \sim \chi_{n-1}(\phi_2)$$

$$2. \chi_n(\phi_1) = \chi_1(\phi_1) \rightarrow \chi_2(\phi_1)$$

Consider:

$\vdots$

$$k. \chi_1(\phi_1) \rightarrow \chi_1(\phi_2) \quad \text{by IH}$$

$$k+1. \chi_2(\phi_1) \rightarrow \chi_2(\phi_2) \quad \text{by IH}$$

$\vdots$

$$k+j. \chi_1(\phi_2) \rightarrow \chi_1(\phi_1)$$

$$3. \chi_n(\phi_1) = \Box \chi_{n-1}(\phi_1)$$

Then consider:

$$k. \chi_{n-1}(\phi_1) \rightarrow \chi_{n-1}(\phi_2) \quad \text{by IH}$$

$$k+1. \Box \chi_{n-1}(\phi_1) \rightarrow \Box \chi_{n-1}(\phi_2) \quad \text{Boxler}$$

$\vdots$

$$k+j. \chi_{n-1}(\phi_2) \rightarrow \chi_{n-1}(\phi_1) \quad \text{by IH}$$

$$k+j+1. \Box \chi_{n-1}(\phi_2) \rightarrow \Box \chi_{n-1}(\phi_1) \quad \text{Boxler}$$

$$k+j+2. \Box \chi_{n-1}(\phi_1) \leftrightarrow \Box \chi_{n-1}(\phi_2) \quad \text{by def}$$

$$\therefore \vdash_k \Box \chi_{n-1}(\phi_1) \leftrightarrow \Box \chi_{n-1}(\phi_2)$$

Q4. Our new definition of $\Gamma \vdash_K \phi$ is that a wff ϕ is provable from Γ iff $\vdash_K (\gamma_1 \wedge \dots \wedge \gamma_n) \rightarrow \phi$ and $\gamma_1, \dots, \gamma_n \in \Gamma$
 or $\Gamma = \emptyset$ and $\vdash_K \phi$

For this question, we will establish the following facts about inconsistent sets.

Lemma: Let $\Gamma \vdash_S$ denote that $\Gamma \vdash_S \sim(P \rightarrow P)$ i.e. Γ is inconsistent. Then

(L1) $\Gamma \vdash_S$ iff for every ϕ , $\Gamma \vdash_S \phi$

(L2) $\Gamma, \sim\phi \vdash_S$ iff $\Gamma \vdash_S \phi$

(L3) $\Gamma, \phi \vdash_S$ iff $\Gamma \vdash_S \sim\phi$

Proof of Lemma.

(L1) ($\Rightarrow$) Assume $\Gamma \vdash_S$

We know that $\vdash_S \sim\phi \rightarrow (\phi \rightarrow \psi)$ (MPL-logic)

Then, rewrite as $\vdash_S \sim(\phi \rightarrow \phi) \rightarrow ((\phi \rightarrow \phi) \rightarrow \psi)$

By definition of provability from a set,

$\sim(\phi \rightarrow \phi) \vdash_S (\phi \rightarrow \phi) \rightarrow \psi$

We also know that $\vdash_S \phi \rightarrow \phi$ (MPL-logic)

By Cut, $\sim(\phi \rightarrow \phi) \vdash \psi$

by Cut again, $\Gamma \vdash \psi$ for any ψ

($\Leftarrow$) for any ϕ , $\Gamma \vdash_S \phi$

Trivially, we let $\phi = \sim(P \rightarrow P)$

$\therefore \Gamma \vdash_S \text{ } \times$

You have not
shown that our
new definition of
 $\vdash \phi$

(L2) ($\Rightarrow$) Assume $\Gamma, \sim\phi \vdash_S$

By def of provability $\rightarrow$ this does not follow from the definition. It's true, but there is a bit of work to get there

We want to show that

$$\sim\phi \rightarrow \sim(\phi \rightarrow \phi) \vdash_S \phi$$

$$1. \sim\phi \rightarrow \sim(\phi \rightarrow \phi) \text{ Premise}$$

$$2. (\phi \rightarrow \phi) \rightarrow \phi \text{ PL (contradiction)}$$

$$3. (\phi \rightarrow \phi) \text{ PL then}$$

$$4. \phi$$

By Cut, $\Gamma \vdash_S \phi$

$\rightarrow$ & again, you haven't shown that cut works

($\Leftarrow$) Assume $\Gamma \vdash_S \phi$

We want that $\phi, \sim\phi \vdash_S$

$$1. \phi \text{ premise}$$

$$2. \sim\phi \text{ premise}$$

$$3. \phi \rightarrow (\sim(\phi \rightarrow \phi) \rightarrow \phi) \text{ PL1}$$

$$4. \sim(\phi \rightarrow \phi) \rightarrow \phi \text{ MP, 1, 3}$$

$$5. \sim\phi \rightarrow (\sim(\phi \rightarrow \phi) \rightarrow \sim\phi) \text{ PL1}$$

$$6. \sim(\phi \rightarrow \phi) \rightarrow \sim\phi \text{ MP, 2, 5}$$

$$7. (\sim(\phi \rightarrow \phi) \rightarrow \sim\phi) \rightarrow ((\sim(\phi \rightarrow \phi) \rightarrow \phi) \rightarrow \sim(\phi \rightarrow \phi)) \text{ PL3}$$

$$8. (\sim(\phi \rightarrow \phi) \rightarrow \phi) \rightarrow \sim(\phi \rightarrow \phi) \text{ MP, 6, 7}$$

$$9. \sim(\phi \rightarrow \phi) \text{ MP, 5, 8}$$

By Cut, $\Gamma, \sim\phi \vdash_S \phi$ ~~PL~~ ~~S~~ ~~MP~~

(L3) ($\Rightarrow$) Assume $\top, \phi \vdash_s$

By def of provability,

Same comments here

$$\top \vdash_s \phi \rightarrow \neg(\phi \rightarrow \phi)$$

By composition,

$$\top \vdash_s (\phi \rightarrow \phi) \rightarrow \neg\phi$$

We know that $\phi \rightarrow \phi$ is an MPL-tautology

$$\therefore \vdash_s \phi \rightarrow \phi$$

By Cut, $\top \vdash_s \neg\phi$

($\Leftarrow$) Assume $\top \vdash_s \neg\phi$

As in L2,

$$\phi, \neg\phi \vdash_s$$

But Cut

$$\top, \phi \vdash_s \text{ \# }$$

(a) Claim: $\phi \in \ominus$ iff $\ominus \vdash_k \phi$

Proof.

($\Rightarrow$) Assume $\phi \in \ominus$

Then, consider a proof from $\ominus$

$$\ominus \vdash_s \phi \text{ since}$$

1. ϕ premise i.e. ϕ is a member of $\ominus$

($\Leftarrow$) Assume $\ominus \vdash_s \phi$

from the definition

For any ϕ , either $\phi \in \ominus$ or $\neg\phi \in \ominus$

We want to show that given assumption, $\sim\phi \notin \ominus$

AFC that $\sim\phi \in \ominus$.

Let us define $\ominus = T \cup \{\sim\phi\}$

Then, by assumption

$$T, \sim\phi \vdash_S \phi$$

Then, we can do a proof from $T, \sim\phi$

1. ϕ By assumption

2. $\sim\phi$ Premise

3. $\phi \wedge \sim\phi$ def of $\wedge$

4. $\sim(\phi \rightarrow \phi)$ def of $\rightarrow$

Then $\ominus \vdash_S \sim(\phi \rightarrow \phi)$

But $\ominus \not\vdash_S \text{ } \times$

So, $\sim\phi \notin \ominus$. By def of $\ominus$,

$\phi \in \ominus$ ~~$\times$~~

(b) i. Claim: $\sim\phi \in \ominus$ iff $\phi \notin \ominus$

Proof. ($\Rightarrow$) Assume $\sim\phi \in \ominus$

AFC that $\phi \in \ominus$

Then, Consider a proof from $\ominus$

1. ϕ premise

2. $\sim\phi$ premise

3. $\phi \wedge \sim\phi$ PL

4. $\sim(\phi \rightarrow \phi)$ def of $\rightarrow$

So, $\ominus \vdash_S \sim(\phi \rightarrow \phi) \rightarrow$

But by def of $\ominus$, $\ominus \not\vdash_S \sim(\phi \rightarrow \phi)$ ~~$\times$~~ you use $\sim(\phi \rightarrow \phi)$ for the contradiction when you should use $\sim(\phi \rightarrow \phi)$

So, $\phi \notin \ominus$

($\Leftarrow$) Assume $\phi \notin \ominus$,

By def of $\ominus$, for any wff ϕ

either $\phi \in \ominus$ or $\sim\phi \in \ominus$

✓

Then, $\sim\phi \in \ominus$ ✗

ii. Claim: $\phi \rightarrow \psi \in \ominus$ iff $\phi \notin \ominus$ or $\psi \in \ominus$

Proof: ($\Rightarrow$) Assume $\phi \rightarrow \psi \in \ominus$

Then $\ominus \vdash_s \phi \rightarrow \psi$

Since

1. $\phi \rightarrow \psi$ Premise

AFC that $\phi \in \ominus$ and $\psi \notin \ominus$

By the definition of $\ominus$,

$\sim\psi \in \ominus$

So, $\{\phi, \sim\psi\} \subset \ominus$

Now, We want to show that, given the assumption,

$\ominus \vdash_s \sim(\phi \rightarrow \psi)$

1. $\sim\psi$ Premise

2. ϕ Premise

3. $\sim(\phi \rightarrow \psi)$ PL

Since we can obtain a proof of $\phi \rightarrow \psi$ & a proof of $\sim(\phi \rightarrow \psi)$ from $\ominus$
consider

⋮

k. $\phi \rightarrow \psi$

⋮

k+j. $\sim(\phi \rightarrow \psi)$

k+j+1. $(\phi \rightarrow \psi) \wedge \sim(\phi \rightarrow \psi)$

k+j+2. $\sim(\phi \rightarrow \psi) \rightarrow (\phi \rightarrow \psi)$

that means that
 $\neg t_s$ Yes, but again you use the
wrong contradiction

But the def of $\neg$ implies

$$\neg t_s \text{ :X:}$$

$$\text{So, } \phi \notin \neg \text{ or } \psi \in \neg$$

($\Leftarrow$) Assume $\phi \notin \neg$ or $\psi \in \neg$

Then we have two cases:

$$1) \quad \phi \notin \neg \text{ implies } \sim \phi \in \neg$$

It follows that

$$\neg t_s \phi \rightarrow \psi$$

Since

$$1. \sim \phi \text{ Premise}$$

$$2. \sim(\phi \rightarrow \psi) \rightarrow \phi \quad \text{PL}$$

$$3. \sim \phi \rightarrow \sim(\phi \rightarrow \psi) \quad \text{PL}$$

$$3. \sim \sim(\phi \rightarrow \psi) \quad \text{MP 1, 3}$$

$$4. \phi \rightarrow \psi \quad \text{PL}$$

$$\text{From (a) since } \neg t_s \phi \text{ iff } \phi \in \neg,$$

$$\phi \rightarrow \psi \in \neg$$

$$2) \quad \psi \in \neg$$

It follows that

$$\neg t_s \phi \rightarrow \psi$$

$$1. \psi \text{ Premise}$$

$$2. \psi \rightarrow (\phi \rightarrow \psi) \quad \text{PL 1}$$

$$3. \phi \rightarrow \psi \quad \text{MP 1, 2}$$

$$\text{By (a), } \phi \rightarrow \psi \in \neg \text{ :X:}$$

$$\phi \rightarrow \psi \in \ominus \text{ iff } \phi \notin \ominus \text{ or } \psi \in \ominus$$

ii. Claim: $\Box \phi \in \ominus$ iff, for every maximally consistent set Σ st. $R\ominus \Sigma$, $\phi \in \Sigma$

Proof. ($\Rightarrow$) Assume $\Box \phi \in \ominus$

Then, for some arbitrary Σ where $R\ominus \Sigma$,

$$\text{since } \{\phi : \Box \phi \in \ominus\} \subseteq \Sigma$$

This means that $\{\phi\} \subseteq \Sigma$

$$\therefore \phi \in \Sigma$$

($\Leftarrow$) Assume for every maximally consistent set Σ st. $R\ominus \Sigma$, $\phi \in \Sigma$

Assume for contradiction that $\Box \phi \notin \ominus$

By def of $\ominus$ being S-consistent,

$$\neg \Box \phi \in \ominus$$

$\downarrow$ why does this follow?

$\therefore$ there is some Σ st. $R\ominus \Sigma$ and $\phi \notin \Sigma$

which contradicts our assumption